Course Name: Advanced Programming (OOP) Lab

Project Title: Multiplayer Turn-Based Game Engine using Java

Student Name: Subham Pal

Roll Number: 82

Instructor Name: Soumadip Biswas

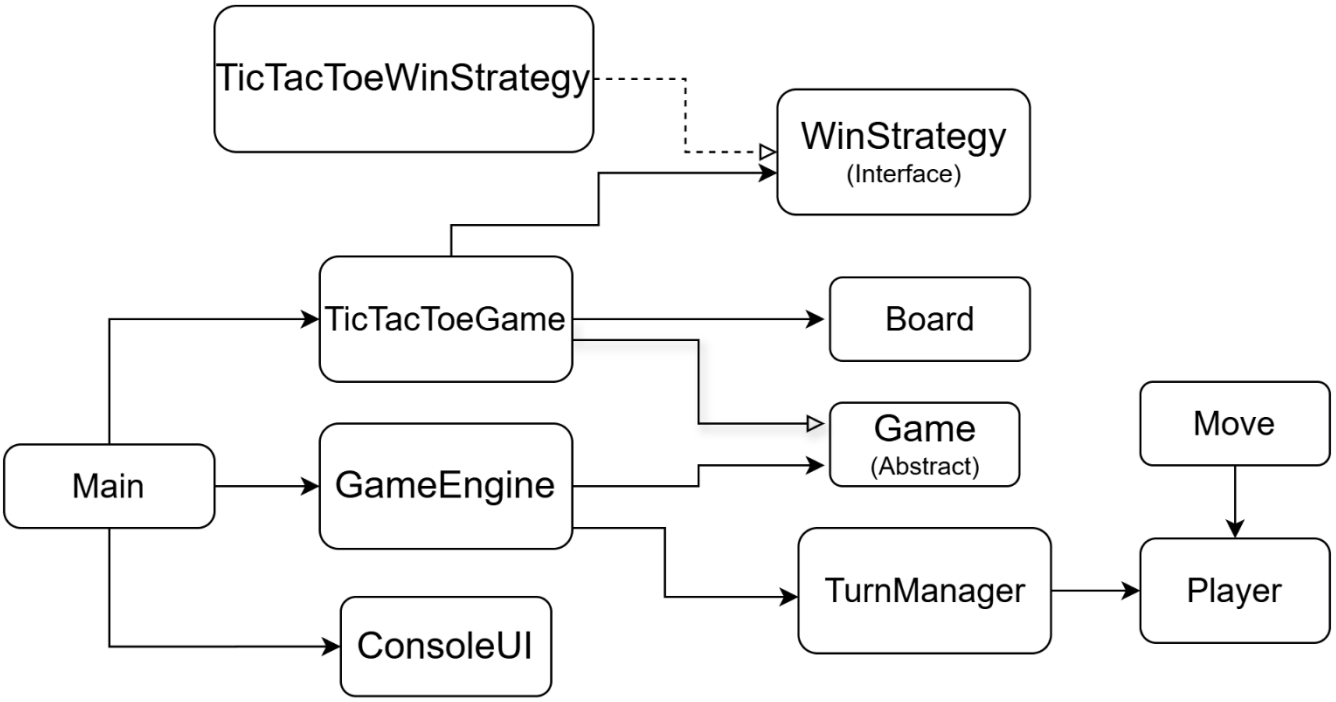
Spring 2026

1. PROBLEM STATEMENT:

Code reuse is challenging in many turn-based games because game logic is closely tied to particular rules. This project creates a reusable Java Turn-Based Game Engine that distinguishes between game-specific logic and game flow (players, turns, state management). The Strategy Pattern and object-oriented concepts like abstraction, inheritance, and polymorphism make it simple to extend the system to support a variety of games. The objective is to develop an architecture that is extensible, maintainable, and modular.

2. SYSTEM DESIGN:

(A) UML CLASS DIAGRAM



(B) DESCRIPTION OF MAJOR CLASSES:

- *Game (Abstract Class): Establishes the framework for every game*
- *TicTacToeGame: Uses logic from Tic-Tac-Toe*
- *GameEngine: Manages the flow of game execution*
- *TurnManager: Manages player turns*
- *Player: Keeps player information*

- *Move: Indicates a movement*
- *Board: Upholds the game grid*
- *WinStrategy: Winner logic interface*
- *ConsoleUI: Manages user communication*

(C) PACKAGE STRUCTURE:

app/

engine/

model/

games/tictactoe/

strategy/

ui/

exception/

(D) INTERACTION FLOW:

User → ConsoleUI → GameEngine → Game → Strategy → Result

3. OOP CONCEPTS DEMONSTRATED:

- **Abstraction**

```
public abstract class Game {
   public abstract void makeMove(Move move);
}
```

Where used: Base class for all games
Why used: To define common structure
Explanation: Ensures all games implement move logic

- **Inheritance**

```
public class TicTacToeGame extends Game
```

Where used: Game specialization
Why used: Code reuse
Explanation: Extends base functionality

- **Polymorphism**

game.checkWinner();

Where used: Method overriding
Why used: Different logic for different games
Explanation: Same method, different behaviour

- **Exception Handling**

```
if (!board.isCellEmpty(row, col)) {
   throw new InvalidMoveException("Cell occupied");
}
```

Where used: Move validation
Why used: Prevent invalid actions
Explanation: Ensures safe execution

- **Collections**

List<Player> players = new ArrayList<>();

Where used: Store players
Why used: Dynamic data handling
Explanation: Allows flexible player list

4. IMPLEMENTATION HIGHLIGHTS

Snippet 1: Move Validation

```
if (row < 0 || row >= 3) {
    throw new InvalidMoveException("Invalid position");
}
```

Prevents invalid input

Snippet 2: Strategy Pattern

```
Player winner =
winStrategy.checkWinner(board.getGrid());
```

Separates rules from logic

Snippet 3: Draw Detection

```
if (isBoardFull()) {
    gameState = GameState.DRAW;
}
```

Handles game end condition

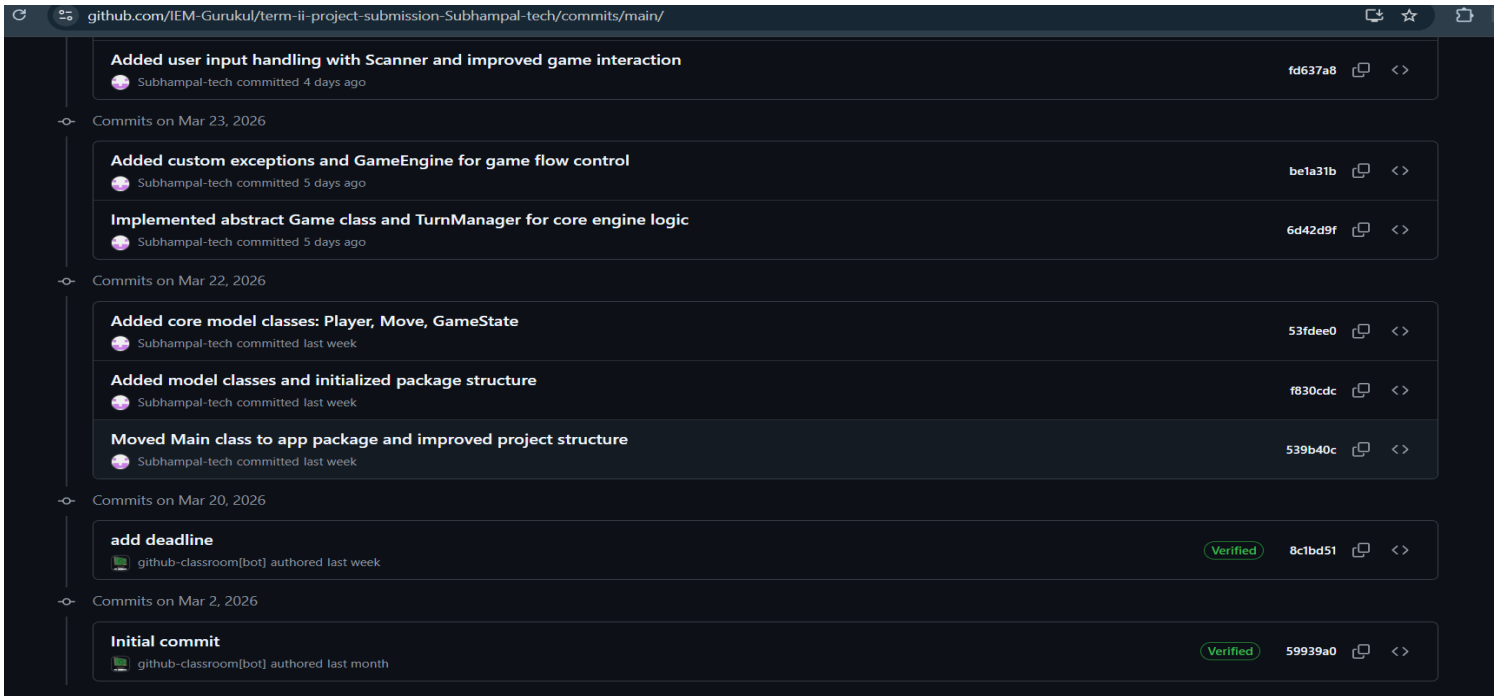
5. TESTING & ERROR HANDLING

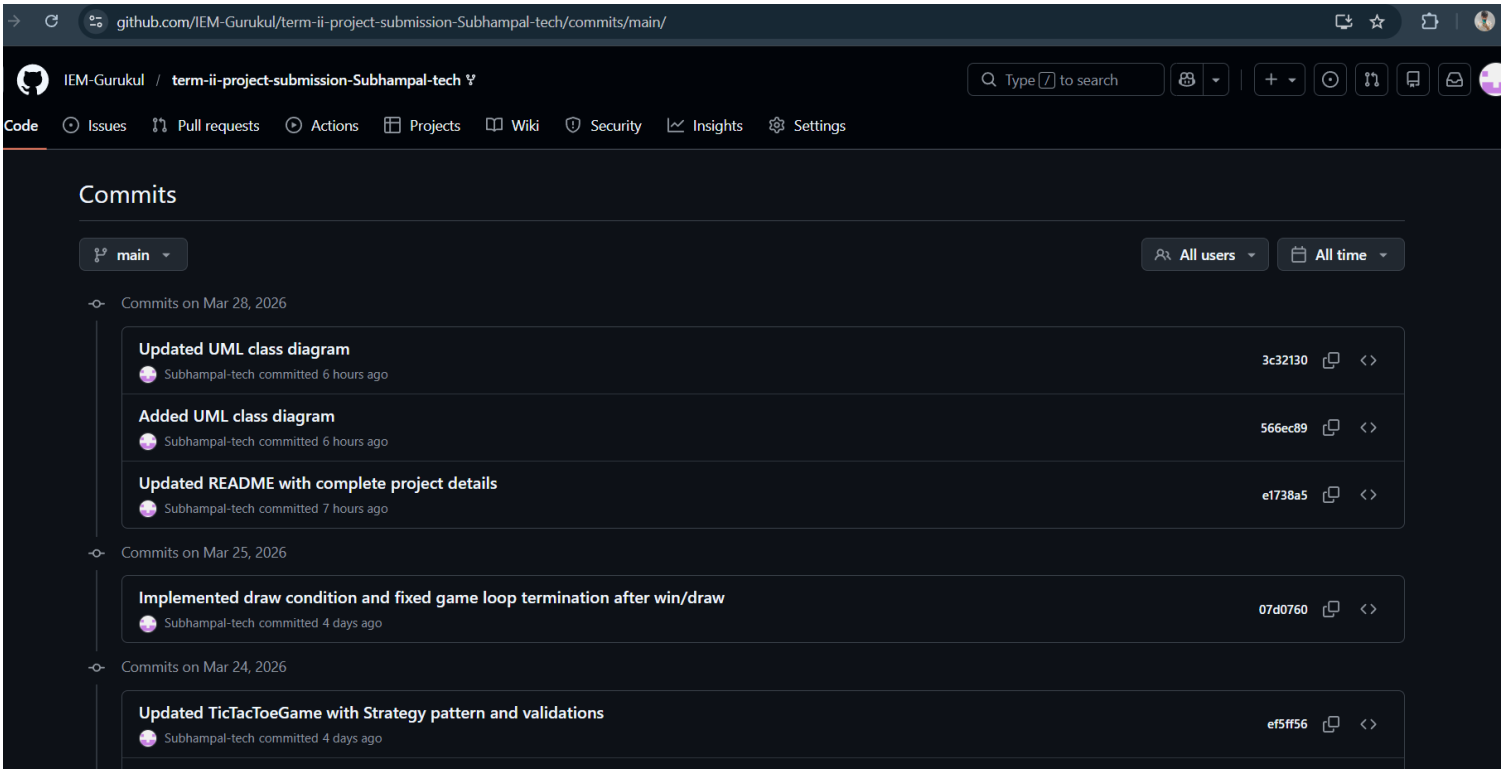
- **Test Cases**
  - *Valid move*
  - *Invalid move*
  - *Same cell*
  - *Win condition*
  - *Draw condition*

- **Edge Cases**
  - *Input out of range*
  - *Full board*
  - *Repeated moves*

- **Failure Handling**
  - *Custom exceptions*
  - *Input validation*
  - *Graceful messages*

6. GIT WORKFLOW





With several significant commits, the project was developed incrementally. Project structure and core class creation were among the first commits. Game logic, design pattern implementation, and UI integration were added in later commits. Bug fixes, input validation, and documentation were the main topics of final commits.

7. CONCLUSION & FUTURE SCOPE

Conclusion:

The project effectively uses OOP concepts to demonstrate a modular and reusable turn-based game engine.

Future Scope:

- *Add GUI*
- *Add multiplayer networking*
- *Add AI opponent*

8. APPENDIX

Appendix A: Initial Project Proposal

Project Title: Multiplayer Turn-Based Game Engine with Java
Problem Statement: Many simple games, such as Tic-Tac-Toe, card games, and strategy board games, have a turn-based structure in which players perform actions sequentially. In most implementations, the game logic and rules are inextricably linked to one game, making it difficult to reuse the code for another game. The goal of this project is to create a tiny Turn-Based Game Engine in Java that handles players, turns, and game data independently of the actual game rules. This engine will allow alternative games to be developed by extending the base classes and creating their own rules. The project focuses on demonstrating good object-oriented architecture, modular structure, and error handling, while maintaining The system is easy to use and expandable.
Target User: Students or novice developers who wish to learn how object-oriented programming principles can be used to organize turn-based multiplayer games.
Core Features:

- Add and oversee several players
- Use the turn management system to decide who gets to play next.
- A depiction of player movements in the game
- Management of game states, including continuing, completed, and winner detection
- An example of how to use the engine to construct a basic game (Tic-Tac-Toe)
- Using exception handling to deal with invalid moves
- Using collections to keep moves and players

OOP Concepts to be Used:

- **Abstraction:** A turn-based game's overall behaviour will be defined by an abstract Game class.
- **Inheritance:** Certain games, such as Tic-Tac-Toe Game, will add their own rules to the underlying Game class.
- **Polymorphism:** Various game implementations will supersede techniques like move validation and winner checking.
- **Exception Handling:** Cases like invalid movements or attempts to play after the game has ended will be handled by custom exceptions
- **Collections:** Players and history will be stored in Java collections such as Array List.

Proposed Architecture Description:
The system will be designed using a modular, object-oriented architecture. By coordinating players, turns, and game state, a central game engine will manage the game's flow. Any turn-based game will have a common structure defined by a base abstract Game class. This class will be expanded with particular games, which will apply their own logic and rules. Player rotation will be handled by a Turn Manager class, while individual classes like Player and Move will represent participants and their actions. This design preserves the engine's reusability and makes it simple to add additional games without altering the fundamental framework.

Appendix B: Faculty Approval Mail

PCCCS495 Proposal Review – APPROVED_MINOR

S soumadip.biswas@ieminternational.ai

To: Subham Pal

Tue 3/10/2026 9:05 AM

Dear Subham Pal,

Project: Multiplayer Turn-Based Game Engine using Java
Decision: APPROVED_MINOR
Score: 92

Strengths:
The proposal clearly identifies a common problem in game development regarding code reusability and effectively proposes an OOP-centric solution. The problem statement, target audience, and feature list are well-defined and realistic for a university project. The explicit mention of core OOP concepts (Abstraction, Inheritance, Polymorphism, Encapsulation) and a modular architecture centered around an abstract `Game` class and specific game implementations (like Tic-Tac-Toe) demonstrates a strong understanding of how to achieve the stated goals. The separation of game logic from game flow is a key strength, promising a robust and extensible engine. The inclusion of exception handling and a clear example application (Tic-Tac-Toe) further solidifies the proposal's quality.

Improvements:
While the proposal clearly outlines a 'Multiplayer Turn-Based Game Engine,' it would benefit from a brief clarification on the intended scope of 'multiplayer.' Specifically, will it focus on local multiplayer (e.g., two players sharing a single instance of the game) or does it intend to explore basic network multiplayer (e.g., using sockets)? For a university project, clarifying this will help manage expectations and scope. Additionally, while 'Design Pattern' is listed as an OOP concept, it would strengthen the proposal to briefly mention one or two specific design patterns that are likely to be employed (e.g., Strategy pattern for game rules, Observer pattern for game state changes) and how they might contribute to the architecture.

Regards,
Course Instructor